AWS from Zero

Account & IAM Setup - Console (GUI) and CLI

A complete, click-by-click guide for absolute beginners

AI Engineer Essentials - Pre-class Preparation

Contents

Before You Begin

Who this is for: anyone who has never used AWS. By the end you will have an AWS account, a secure day-to-day login, and a working command line - everything needed to deploy the course project.

Time required: about 30-45 minutes. Do this **before** class so we can start building immediately.

What you need: an email address, a phone number, and a credit/debit card (AWS requires a card to verify your identity; the course stays on free / low-cost tiers).

> THE BIG PICTURE
>
> **The one-sentence mental model.** You create one all-powerful **root** account (like the master key), immediately lock it away, and do all of your daily work through a safer **IAM user** - which can sign in to the website (the **Console / GUI**) and run commands from your terminal (the **CLI**).

Part 0 - Five Words You Must Know

Skim this once; it makes every later step obvious.

Term	What it means in plain English
Root account	The original owner of the AWS account, created with your email. It can do absolutely anything - including delete the whole account and change billing. Because it is so powerful, you use it almost never.
IAM user	A separate, named login you create for everyday work (e.g. your name, or ‘nawaloka-agent’). You give it only the permissions it needs. This is what you sign in with day to day.
Console (GUI)	The AWS website you click around in (console.aws.amazon.com). You log in with a username + password.
CLI	The ‘Command Line Interface’ - you type commands in your terminal (e.g. aws s3 ls). It logs in with an access key, not a password.
Region	Which physical part of the world your resources live in. This course uses us-west-2 (Oregon). Always check the region selector (top-right of the console).

> DON'T MIX THESE UP
>
> **Password vs Access Key.** The Console uses a **password** (something you type). The CLI uses an **access key** (a long ID + secret that programs use). They are two different credentials for the **same** IAM user - you will create both.

Part 1 - Create Your AWS Account

Skip to Part 2 if you already have an AWS account.

1. Open **https://aws.amazon.com/** and click **“Create an AWS Account”** (top-right).
2. Enter your **email address**, choose an **AWS account name** (e.g. “Your Name - AEE”), and verify the email with the 6-digit code AWS sends you.
3. Set the **root user password**. Use a strong, unique one and save it in a password manager.
4. Choose account type **Personal**, then fill in your name, phone, and address.
5. Add a **payment method** (credit/debit card). AWS may place a temporary ~$1 authorisation to verify the card - it is refunded.
6. Complete **phone/SMS verification** (enter the code AWS calls or texts you).
7. Choose the **Basic support plan - Free**.
8. Click **“Go to the AWS Management Console.”** Your account is live.

> REASSURANCE
>
> **Card safety.** You will not be charged for simply having an account. Real charges come only from resources you create - and Part 8 adds a budget alarm so nothing surprises you.

Part 2 - Lock Down the Root Account (MFA)

Why: if someone steals your root login they own everything. Multi-Factor Authentication (MFA) means a thief also needs your phone. Do this now.

1. Sign in as **root** (the email you just used). In the top search bar type `IAM` and open the **IAM** console.
2. On the IAM dashboard you'll see a **“Security recommendations”** box. Click **“Add MFA”** for the root user. (Or: top-right account menu → **Security credentials**.)
3. Click **“Assign MFA device”**, choose **Authenticator app** (e.g. Google Authenticator, Authy, 1Password).
4. Scan the QR code with your phone app, type two consecutive 6-digit codes, and click **“Add MFA.”**
5. Done. From now on, signing in as root requires the code from your phone.

> REMEMBER
>
> **Golden rule.** After this, **do not use root again** except for the rare account-level tasks this guide calls out. All real work happens as the IAM user you create next.

Part 3 - Turn On Billing Visibility (Root Only)

Why this matters: by default, IAM users - **even admins** - get "Access Denied" on the Billing and Cost pages. This one switch fixes it, and only **root** can flip it. Do it now while you're signed in as root.

1. As **root**, click your **account name** (top-right) → **Account**.
2. Scroll to **"IAM user and role access to Billing information."** Click **Edit**.
3. Tick **"Activate IAM Access"** and click **Update**.

> **COMMON GOTCHA**
> **This is the fix for "I can't see the Cost dashboard."** Without it, your IAM user's billing pages are blank no matter what permissions it has.

Part 4 - Create Your IAM User (Console + CLI)

This creates the login you'll actually use. We give it **both** a Console password (for the website) and an access key (for the terminal).

4A · Create the user and Console (GUI) access

1. In the search bar open **IAM** → left menu **Users** → **"Create user."**
2. Enter a **User name** (e.g. `nawaloka-agent` or your own name).
3. Tick **"Provide user access to the AWS Management Console."**
   a. Choose **"I want to create an IAM user."**
   b. Console password: pick **"Custom password"** and set one you'll remember (or use Autogenerated).
   c. Optionally **untick** "Users must create a new password at next sign-in" to keep it simple.
4. Click **Next** to reach the permissions screen (continued in 4B).

4B · Give it permissions

1. On "Set permissions," choose **"Attach policies directly."**
2. Search for **AdministratorAccess** and tick it.
3. Click **Next** → **"Create user."**

> **TEACHING NOTE**
> **About AdministratorAccess.** It grants full power - perfect for learning so nothing blocks you. In a real job you'd grant only what's needed (least privilege). For this course it's a deliberate, accepted simplification.

4C · Create the CLI access key

1. Open the user you just created → **"Security credentials"** tab.
2. Scroll to **"Access keys"** → **"Create access key."**
3. Choose use case **"Command Line Interface (CLI),"** tick the confirmation box, click **Next** then **"Create access key."**

4. Click **“Download .csv file.”** This holds the **Access key ID** and **Secret access key** - you'll need them in Part 6.

CRITICAL - SECRETS

You see the secret key ONCE. Download the .csv now and keep it safe. If you lose it, you simply delete that key and create a new one - no harm done.

Never share or commit it. Treat access keys like passwords: never paste them into chat, email, screenshots, or Git.

Part 5 - Sign In as Your IAM User (Console / GUI)

IAM users sign in at a different URL than root. Let's make it friendly and bookmark it.

1. (Optional, nicer URL) As root or admin, open **IAM** → dashboard. Under **"AWS Account"** find **"Account Alias"** → **Create** → set something memorable like `aee-yourname`.
2. Your IAM sign-in URL is now **https://<alias>.signin.aws.amazon.com/console** (or https://<account-id>.signin.aws.amazon.com/console). **Bookmark it.**
3. Sign out of root. Open that URL and log in with your **IAM user name + Console password** from Part 4A.
4. Top-right, set the **Region** selector to **"US West (Oregon) us-west-2."**

> QUICK CHECK
> **How to tell who you're logged in as.** Click the top-right name. Root shows your account email; an IAM user shows `username @ account-id`. For daily work it should always be the IAM user.

Part 6 - Install & Configure the AWS CLI

6A · Install the CLI

macOS (with Homebrew):

```
brew install awscli
```

macOS / Windows / Linux (official installer): follow https://aws.amazon.com/cli/ - download and run the installer for your OS.

Verify it's installed:

```
aws --version
```

You should see something like `aws-cli/2.x.x ...`.

6B · Configure your credentials

Run this and paste the values from the .csv you downloaded in Part 4C:

```
aws configure --profile nawaloka
```

It will ask four things:

- **AWS Access Key ID** - from the .csv
- **AWS Secret Access Key** - from the .csv
- **Default region name** - type `us-west-2`
- **Default output format** - type `json`

> TIP
> **Why "--profile nawaloka"?** A profile is a named set of credentials. Using one keeps this course's account separate from any other AWS account you might add later. You then add `--profile nawaloka` to commands, or set `export AWS_PROFILE=nawaloka` once per terminal.

Part 7 - Verify Everything Works

Confirm the CLI is talking to AWS as your IAM user:

```
aws sts get-caller-identity --profile nawaloka
```

Expected output (your IDs will differ):

```
{
  "UserId": "AIDA....",
  "Account": "310485117018",
  "Arn": "arn:aws:iam::310485117018:user/nawaloka-agent"
}
```

If the `Arn` ends in **user/<your-iam-name>**, you're done - the CLI is authenticated. (If it says `:root`, you configured root keys by mistake - delete them and use the IAM user's keys.)

Part 8 - Add a Cost Guardrail (Budget Alert)

Why: a forgotten resource is the #1 way beginners get a surprise bill. A budget emails you the moment spend crosses a threshold.

1. In the Console search bar open **"Billing and Cost Management"** → left menu **Budgets** → **"Create budget."**
2. Choose **"Cost budget"**, click **Next**.
3. Set a **monthly amount** (e.g. `$20`), give it a name.
4. Add an **alert threshold** (e.g. at 80% and 100%) and your **email**.
5. Create the budget. You'll now get an email if spend approaches the limit.

CLI users can confirm budgets exist with:

```
aws budgets describe-budgets --account-id 310485117018 --profile nawaloka
```

Part 9 - Security Habits (Keep These Forever)

- **Never use root for daily work** - only the rare account/billing tasks flagged in this guide.
- **MFA on everything** - root and, ideally, your IAM user too (IAM → your user → Security credentials → Assign MFA).
- **Never commit or share access keys** - keep them out of Git (use a .gitignore), chat, and screenshots.
- **Rotate and delete keys** - if a key leaks, delete it immediately (IAM → user → Security credentials) and create a new one.
- **One profile per account** - keeps credentials tidy and prevents deploying to the wrong place.
- **Always check the Region** - resources in the wrong region look 'missing' and cost money quietly.
- **Turn things off when done** - stop or delete resources after each session (see the cost document).

Part 10 - Troubleshooting Common Errors

Symptom	Cause & fix
"Access Denied" on Billing / Cost pages	The root-only billing toggle is off. Sign in as root → Account → activate "IAM user and role access to Billing information" (Part 3).
Cost dashboard looks empty	Billing data lags ~24h, and Cost Explorer takes ~24h to first populate. Also costs may simply be ~$0 so far. Wait a day.
"Unable to locate credentials" (CLI)	You haven't run aws configure for this profile, or you forgot --profile nawaloka. Re-run Part 6B.
get-caller-identity shows :root	You configured root access keys. Delete them (root → Security credentials), create an IAM user key (Part 4C), reconfigure.
"InvalidClientTokenId" / "SignatureDoesNotMatch"	The access key is wrong, was deleted, or has a typo. Create a fresh key (Part 4C) and run aws configure again.
Resources you created aren't showing	Wrong Region. Set the top-right selector to us-west-2 (and pass --region us-west-2 on the CLI).
Can't sign in as the IAM user	You're using the root URL. Use https://<alias>.signin.aws.amazon.com/console with the IAM user name + Console password (Part 5).

You're Ready

You now have: a secured root account, an MFA-protected setup, an IAM user that signs in to both the **Console** and the **CLI**, billing visibility, and a budget guardrail. That's the full foundation - bring this to class and we'll deploy on top of it.

BRING THIS TO CLASS

Pre-class checklist: (1) IAM user created · (2) can sign in to the Console as that user · (3) `aws sts get-caller-identity` returns your user ARN · (4) Region = us-west-2 · (5) a budget alert exists.